



Antibacterial Activity of Algerian Lichens:
A Promising Source of Novel Antimicrobial Agents against Antibiotic Resistance

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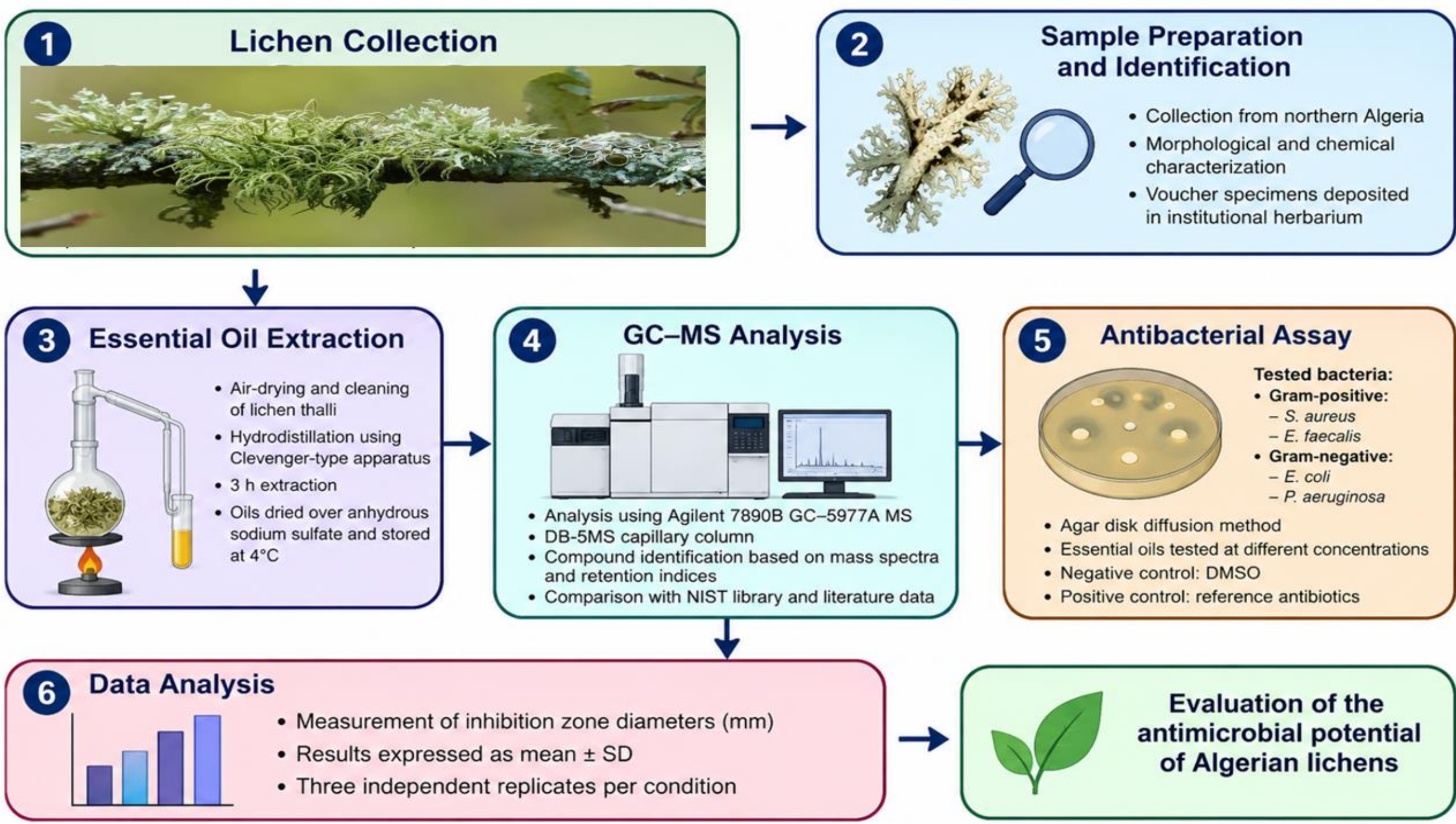
INTRODUCTION

Antimicrobial resistance (AMR) is a major global health challenge, increasing the urgent need for new antimicrobial agents. Lichens are unique symbiotic organisms producing diverse secondary metabolites with important biological activities, including antibacterial properties [1]. Their volatile compounds and specialized metabolites represent promising sources of bioactive molecules. Algerian lichens remain relatively underexplored despite their potential as a source of chemically diverse antimicrobial compounds.

This study aimed to characterize the volatile chemical composition and evaluate the antibacterial activity of four Algerian lichens—*Ramalina pollinaria*, *Pseudevernia furfuracea*, *Evernia prunastri*, and *Ramalina farinacea*—against Gram-positive and Gram-negative bacterial strains, highlighting their potential as natural sources of promising antimicrobial agents.

MATERIALS AND METHODS

- Four Algerian lichens were investigated: *Ramalina pollinaria*, *Pseudevernia furfuracea*, *Evernia prunastri*, and *Ramalina farinacea*.
- Thalli were collected from selected sites in northern Algeria (forests of Babor, and Djebel Megriss) and identified using morphological and chemical characteristics.
- Voucher specimens were deposited in institutional herbaria for reference.
- Lichen material was air-dried, cleaned, and subjected to **hydrodistillation using a Clevenger-type apparatus for 3 h**.
- The obtained essential oils were dried over anhydrous sodium sulfate and stored at **4°C** until analysis.
- Volatile compounds were characterized by **GC–MS** using a DB-5MS capillary column.
- Compound identification was based on **mass spectra and retention indices**, using the NIST spectral library and published reference data.
- Antibacterial activity was evaluated by the **agar disk diffusion method** against Gram-positive and Gram-negative bacterial strains. Essential oils were tested at different concentrations, with appropriate negative and antibiotic controls.
- Inhibition zones (mm ± SD) were determined from three independent replicates after 24 h incubation at 37°C.**



RESULTS

Chemical Composition

GC–MS analysis revealed distinct chemical profiles among the four Algerian lichens. *Ramalina pollinaria* was dominated by manool (34.03%), followed by octadecane (11.59%) and manol-13-epi (11.10%). *Pseudevernia furfuracea* was characterized mainly by manool (21.11%), α-pinene (15.23%), and abietol-4-epi (12.91%). In contrast, *Evernia prunastri* showed a marked predominance of n-octadecanol (54.06%), while *Ramalina farinacea* was mainly characterized by manool (14.21%), n-octadecanol (10.24%), and eicosene-1 (8.26%).

Table 1. Chemical composition

Lichen species	Major compound	Relative abundance (%)
<i>Ramalina pollinaria</i>	Manool	34,03
	Octadecane	11,59
	Manol-13-epi	11,10
	Oleic acid	5,65
<i>Pseudevernia furfuracea</i>	Abietatriene	4,17
	Manool	21,11
	α-Pinene	15,23
	Abietol-4-epi	12,91
<i>Evernia prunastri</i>	Abietol	11,13
	Abietatriene	10,21
	n-Octadecanol	54,06
	n-Tetradecanol	8,27
<i>Ramalina farinacea</i>	Heptadecane	4,70
	Tridecenol acetate	2,05
	n-Hexadecanol	2,32
	Manool	14,21
<i>Evernia prunastri</i>	n-Octadecanol	10,24
	Eicosene-1	8,26
	n-Tetradecanol	7,68
<i>Ramalina farinacea</i>	Manool oxide-13-epi	5,40

Antibacterial activity

The antibacterial activity of lichen essential oils was evaluated by the disk diffusion method against eight bacterial strains, including both Gram-positive and Gram-negative species. The results revealed a clear variation in the inhibition diameters according to the lichen species and dilution factor . At the pure concentration (1/1), lichens displayed strong antibacterial activity, although *P. furfuracea* showed higher mean inhibition zones against most strains. The pure oil of *R. pollinaria* exhibited the largest inhibition zone against *Bacillus cereus* (21.33 ± 0.6 mm), followed by *Escherichia coli* (12.33 ± 0.6 mm) and *E. faecalis* (10.66 ± 0.6 mm). In contrast, *P. furfuracea* pure oil demonstrated the highest activity against *E. coli* (17.33 ± 1.0 mm) and *S. aureus* (16.33 ± 0.33 mm), with a comparable effect on *B. cereus* (19.0 ± 1.0 mm).

The statistical analysis revealed significant differences in bacterial sensitivity to the essential oil (Table 2). *E. coli*, classified in group “a”, was the most sensitive strain, with a mean inhibition diameter of 16.90 mm, followed by *B. cereus* and *S. aureus*, which showed high susceptibility. In contrast, *E. faecalis*, classified in group “h”, was the most resistant bacterium, exhibiting the lowest inhibition diameter (7.41 mm). These results indicate a clear variability in bacterial response to the essential oil. The desirability analysis showed a predicted overall value of 0.2051 (Fig.01), with pure lichen essential oils exhibiting promising antibacterial activity.

Table 2. Sensitivity of bacteria to Lichen essential oil.

Ran	Name	Mean	n	Non-significant ranges
1	E. coli	16.90	63	a
2	B. cereus	16.32	63	b
3	S. aureus	15.87	63	c
4	P. mirabilis	12.79	63	d
5	P. aeruginosa	12.37	63	e
6	K. pneumoniae	8.30	63	f
7	MRSA	8	63	g
8	E. faecalis	7.41	63	h

LSD 0.05 = 0.18158319418

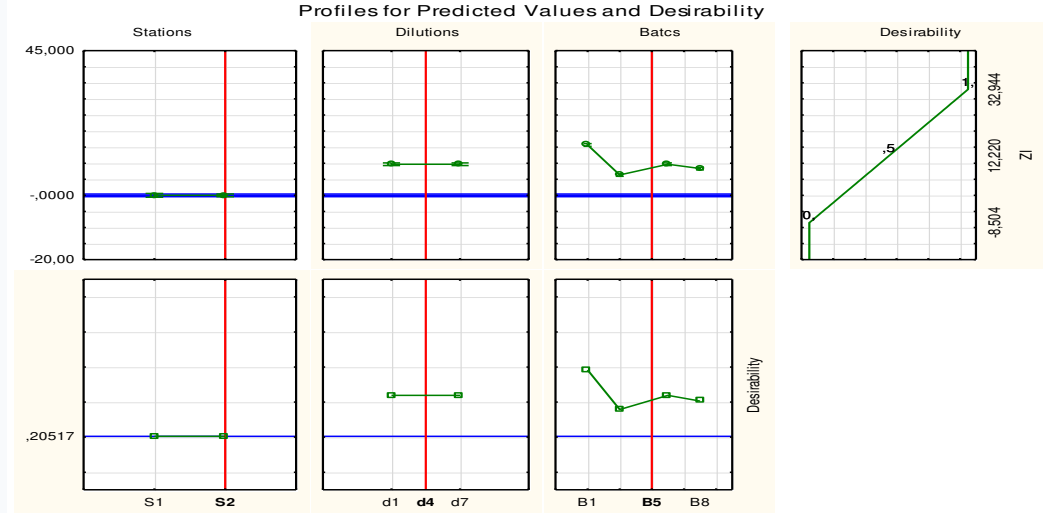


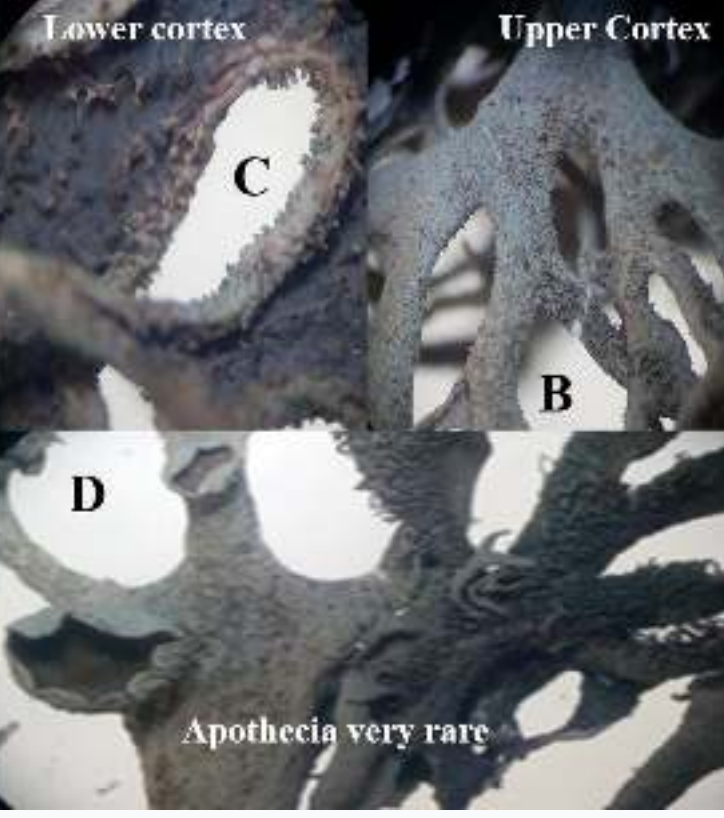
Figure 3. Profile of predicted values and desirability of Lichen essential oil against bacteria tested

5. DISCUSSION

The distinct antibacterial responses observed among the four Algerian lichens appear closely related to their species-specific chemical profiles. *R. pollinaria*, *P. furfuracea*, and *R. farinacea* were characterized by the predominance of manool and diterpenoid compounds, whereas *E. prunastri* was dominated by n-octadecanol, highlighting substantial interspecific chemical variation. Such differences may reflect species-specific biosynthetic pathways and ecological conditions, including substrate and microclimatic factors [2, 3, 4]. Among the investigated species, *P. furfuracea* showed strong activity against *E. coli* and *S. aureus*, while *R. pollinaria* exhibited remarkable inhibition against *B. cereus*. The progressive reduction of inhibition zones following dilution confirmed a clear concentration-dependent antibacterial effect. These findings agree with previous studies reporting important antimicrobial activity of *P. furfuracea* and other lichen extracts [5].

6. CONCLUSION

The four Algerian lichens exhibited distinct chemical profiles and promising antibacterial activity, supporting their potential as natural sources of bioactive compounds. The predominance of manool and abietane-type diterpenoids, particularly in *R. pollinaria* and *P. furfuracea*, may contribute to their inhibitory effects. *P. furfuracea* showed strong activity against *E. coli* and *S. aureus*, while *R. pollinaria* exhibited its highest inhibition against *B. cereus*. The marked reduction in inhibition with increasing dilution confirms a clear concentration-dependent response. Overall, the results highlight the chemical diversity and antibacterial potential of Algerian lichens, warranting further investigation of their active compounds and mechanisms of action.



8. REFERENCES

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